GENESIS
IDEATHON

TEAM & DOMAIN

- Team Name – 4 Frames Per Second
- Team Members – Dhilib SM-CB.EN.U4AEE26009, Gauri Suresh-CB.EN.U4AEE26015, Shyamkrishna Mahesh-CB.EN.U4AEE26052, Sudeesh Kumar-CB.EN.U4AEE26054
- Selected Domain – Energy, Mobility and Smart Infrastructure
- Problem Statement – “Noise pollution: just a disturbance or a source of energy?”



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NOISE DRIVES AWAY LIFE

Noise Waste: Why it's such a big disturbance ♫

- European Environment Agency's 2025 report says" 66,000 premature deaths, 50,000 cases of cardiovascular disease and 22,000 cases of type 2 diabetes- all caused by a single problem: **NOISE** .
- A study by the EEA published in 2026 identifies noise pollution as a trigger for conditions such as depression and as a factor that contributes to reduced quality of life, with several other damaging effects like short term anxiety and irritability leading to long term psychological stress.
- Studies also show disturbed breeding and rearing cycles, psychological stress and habitat abandonment and sometimes even complete extinction of a species (eg; Chinese river dolphins, Chinese paddlefish) all resulting from noise pollution.

AN ACOUSTIC CRISIS TO BE SOLVED

-Noise pollution is one of the most overlooked forms of pollution and with its impact affecting us deeper than we expect, we must be equipped to deal with it effectively with technological, sustainable solutions.

WHO IS AFFECTED?

Target users, stakeholders, beneficiaries ♫

- People in crowded spaces like railway stations, near construction areas and industrial areas, and even more recently, AI data centers that produce a lot of high frequency noise pollution.
- The environment: lots of birds and critters have their breeding and rearing cycles affected and disrupted by man-made noise pollution.
- Low power technologies like IoT, environmental monitoring systems and urban infrastructure stand a lot on improving as their power sources would be changed to a more sustainable one.







CURRENT SITUATION & CONTEXT

Existing scenario, background and how things work today

- **Acoustic energy harvesting** (AEH) is already an active research field. Researchers have already demonstrated systems that convert sound into electricity using **piezoelectric**, **triboelectric**, electromagnetic, and resonator-based approaches.
- But the problem is about the **sound density** i.e. The sound density is very low in public places and the average noise in urban places is about (60 to 80 dB) and because of this low urban sound density the piezoelectric crystal cannot be deformed on its own-which is the main principle in which this conversion lies.
- To overcome this, **efficient resonators** are needed like Helmholtz or Quartz Acoustic Wavelength Resonators which artificially compresses and amplify the generated waves.
- The current workflow of our team focuses on using this piezoelectric technology for noise pollution control and **integrating echo chamber/resonance chamber systems** into the whole circuitry to create a much more efficient and powerful system. Our steps include studying about piezoelectric technology, thinking about creative ways to implement this technology into real world environments.

KEY CHALLENGES

Main drawbacks and consequences ♪

- It can only be used in enclosed areas with a lot of noise (like subways, theatres, long and short tunnels, etc.) to be more efficient and work in optimal conditions.
- Since the echo chambers are still being developed, cost of manufacturing and the exact efficiency of this device used together with the piezoelectric materials (e.g.: quartz or electro-spun membranes) remains in the dark. The materials used for the echo chambers might be too expensive to implement everywhere.
- The existing systems face challenges to convert low frequency noise into usable electrical energy. Energy output depends on sound intensity and frequency, and the conventional designs have less efficiency.

Consequences ♪

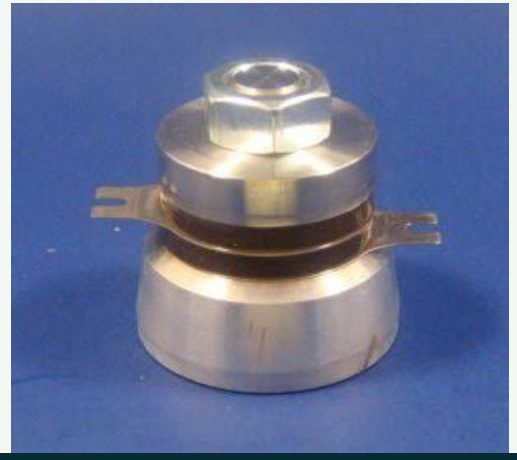
- Without this technology, the **fossil fuel consumption** will still be skyrocketing, this technology can reduce the consumption of these fossil fuels. Also, noise pollution is a real problem, which causes mental and physical health problems, and environmental problems like near extinction of endangered birds and small critters.

EXISTING SOLUTIONS / APPROACHES

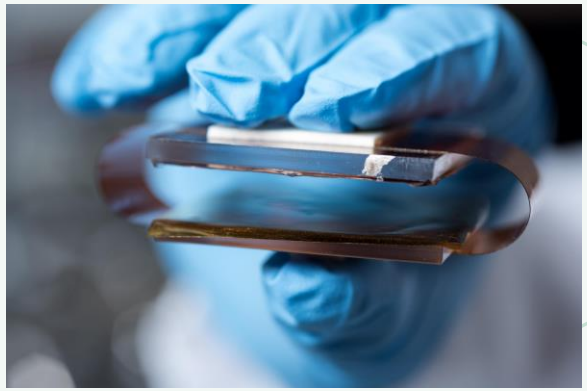
What is already being done, and where it falls short ♪

- The existing methods include piezoelectric and triboelectric devices that convert vibrations made due to the sound created and convert it into electrical energy. **Echo chamber/Resonance chambers** are used to enhance the sound pressure at specific frequencies. These technologies provide a foundation to further develop the acoustic energy harvesting in future.
- Energy output depends on sound intensity and frequency, and the conventional designs may have less efficiency. There is a need for improved resonance-based designs that enhance the sound induced vibrations to produce more electrical energy.
- The triboelectric systems output high voltage and energy densities compared to the piezoelectric counterparts, but the high complexity, maintenance costs and higher cost of these systems make the piezoelectric systems more appealing to mass produce.

Image (a) Piezoelectric transducers:



Image(b) triboelectric transducers:



PROBLEM INSIGHTS & OPPORTUNITY AREAS

Resources required to solve the problem ♪

- Hardware Requirements: Piezoelectric substances, substantial electrical circuitry, acoustic sensors, echo chamber/ resonance chamber constructions
- Software Requirements: Sensor technology
- Other resources: Construction workers and equipment

Opportunities for development:

♫ The technology can be improved by using triboelectric transducers with an altered design as it has better efficiency and output.

♫ This technology is very versatile in its applications. To name a few, it can be used in:

♪placed near car engines to decrease their noise and contribute to power generation by producing additional electrical energy.

♪ placed in construction areas to decrease their cacophony and provide lesser disturbance to surrounding communities.



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INITIAL DIRECTION & EXPECTED OUTCOME

What Is Our Expected Outcome? ♪

- Our team intends to research more on piezoelectric transducers, and their role in a different form of **sustainable energy.**
- Our team would like to improve on the design, piezoelectric technology required, triboelectric devices, more efficient acoustic sensors and cost-effectiveness of the project.
- With our project idea, we should have achieved reduced noise pollution, and a different form of sustainable electricity, more ecologically sustainable society, reduced risk of extinction of birds and critters, reduced risk of hearing loss and injuries in the ear, more comfortable place for the construction and industrial workers who work with high noise equipment and machinery. The people can live healthier and more calmer lives in this high-pressure society.

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TEAM CONTRIBUTION

Team Member	Role	Responsibility in Execution
Dhilib SM	Researcher, Team leader	Existing solutions, Approaches and their Drawbacks.
Gauri Suresh	Researcher, solution analyst, Designer, Ideator	Problem insights and opportunity areas, Introduction, problem analysis
Shyamkrishna Mahesh	Researcher, Fact checker, Creative thinker	Key Challenges, Initial Direction and expected outcome.
Sudeesh Kumar U	Researcher, Strategist,	Current situation & context , Affected demographic